

DRIVER'S MANUAL

# TRUCKHUB

A live instrument cluster for American Truck Simulator & Euro Truck Simulator 2

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TruckHub reads live telemetry from ATS/ETS2's own official SDK - nothing here is a guess or a simulation of a simulation. Every gauge, color, and label in this manual traces back to a real value the game itself is reporting at that instant.

## SECTION 01

# Getting Started

TruckHub is a companion instrument cluster that runs alongside American Truck Simulator or Euro Truck Simulator 2. It doesn't mod the game and doesn't read your screen - it connects to the same official telemetry SDK plenty of racing-sim and trucking dashboards already use, and draws what it gets from that feed as a second, always-on-top card next to your game window.

Launch TruckHub before or after the game - order doesn't matter. The card sits idle with a single line reading *"Waiting for ETS2 / ATS to start..."* until a supported game is actually running, then fills in on its own. Close the game and it goes back to waiting rather than freezing on stale numbers.

## The card itself

Click and drag anywhere on the card's background to move it. The bottom-right corner is a resize handle - drag it to scale the whole card, fonts and all, up or down. Nothing about the card's position or size is saved between component toggles, so if you turn a row off in Settings the card simply gets shorter; turn it back on and it grows again.

Three tabs sit to the right of the card - **GPS**, **LOG**, and **EXT**. Each opens something separate: a live road map, a running job history, and a second window of instruments the main card has no room for. All three are covered in their own sections below.

**TIP** TruckHub stays on top of other windows by design, so you can glance at it without alt-tabbing. If it's in the way, resize it smaller or drag it to a corner - there's no "minimize to tray" here, it's meant to always be visible while you drive.

## Under the hood: CoreBalancer

TruckHub quietly measures real per-core CPU load every minute and steers its own process - and the GPS map's rendering subprocess - away from whichever cores the game is currently loading hardest, instead of guessing a fixed core to avoid. A static "always reserve core 0"-style rule was tried first and, once actually measured against live per-core load, turned out not to match how load really moves around on a real machine while ATS or ETS2 is running.

It also decides how hard the GPS map itself is allowed to work: the map renders at 60fps when the system has headroom, and drops to a steadier 30fps on its own if overall load is high, so a demanding moment in-game - or something else entirely running on your PC - doesn't have to fight the map for the same CPU and GPU time. Changes are deliberately unhurried: a core reassignment only happens when the busiest cores are clearly worse than what TruckHub is already avoiding, and never more than once every five minutes, so ordinary noise in the readings doesn't cause constant switching. None of this needs any configuration - it runs by itself the entire time TruckHub is open.

## SECTION 02

# The Main Dashboard

Top to bottom, exactly the order it appears on the card.

## Title bar

### ☰ Mod Order

Opens a suggested load order for your installed mods, built from community-recommended category tiers. It's a reference list only - TruckHub never touches your Mod Manager, you still reorder mods yourself. See Section 08.

### ⚙ Settings

Opens the two-tab Settings window (Section 07). Glows amber whenever a TruckHub update is available - see the update note in Section 07.

### ⌛ Fullscreen

Toggles the card between its normal size and a larger fullscreen layout.

### ✕ Close

Closes TruckHub. The game keeps running - this only closes the HUD.

## Speed limit sign & turn signals

The sign matches whichever game is running: American Truck Simulator gets the familiar white rectangular **SPEED LIMIT** sign, Euro Truck Simulator 2 gets the red-ringed white circle used across Europe. With no speed limit data for the current road, a dim grey placeholder shows instead of a guessed number.

The two arrows either side of the sign are your actual turn signals, driven straight off the game's own bulb state - they blink at the real rate, and flash together for hazards, because hazards are the same two bulbs turned on at once.

## Lights & controls row

Nine icons, each dim grey when off and lit in its real dashboard color the instant it's active - here's what each one actually is, so you know it on sight the first time it lights up.



### Parking brake

Lit white while the parking brake is set. Off = brake released.



### Retarder

Lit green while engaged - the ring's letter switches to the current level number (1, 2, 3...) instead of staying a plain "R". Only shown on trucks that actually have a retarder fitted.



### Engine brake

Lit green while engaged. Labeled "J" (Jake brake) in ATS, "E" (engine brake) in ETS2 - same feature, the letter just matches what that game's audience calls it.



### Sidelights

A plain lamp, no beam lines. Lit green while your sidelights/parking lights are on.

|                                                                                   |                          |                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Low beam</b>          | Lamp with three beams angled downward. Lit green while low beams are on.                                                                                                                                                             |
|  | <b>High beam</b>         | Lamp with three straight, level beams. Lit blue while high beams are on - the one deliberate exception to the green/amber/red rule (see Section 03).                                                                                 |
|  | <b>Beacon</b>            | A domed light with rays. Lit amber while your roof beacon/hazard light bar is on.                                                                                                                                                    |
|  | <b>Differential lock</b> | A padlock. Lit amber while the diff lock is engaged - it changes how the truck handles, so it's flagged even though nothing's wrong.                                                                                                 |
|  | <b>Lift / tag axle</b>   | A wheel with an arrow above it. Lit amber while the axle is raised; dim while it's down (normal). Only appears at all on a truck that actually has one - see Section 03 for what LIFT/TAG mean on the EXT window's own axle readout. |

## Warning cluster

Seven telltales mirroring the exact warning lights on the game's own dashboard - TruckHub adds no thresholds of its own here, if one of these lights up, the game itself decided it should. Every light in this row goes dark if the truck's electrics are switched off, the same way a real dashboard goes blank with the ignition off.

|                                                                                     |                                  |                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Low air pressure</b>          | The real SAE/ISO air-brake warning symbol - a circled exclamation mark. Red once the air tank pressure drops too low to brake reliably.                                                                                                  |
|  | <b>Emergency / parking brake</b> | The same ring-and-"P" shape as the parking brake status icon above, but this one lives in the warning row and turns red specifically when critically low air pressure has auto-applied the brake for you - not a brake you set yourself. |
|  | <b>Oil pressure</b>              | An oil can with a drip. Red once engine oil pressure drops below safe.                                                                                                                                                                   |
|  | <b>Engine temperature</b>        | A thermometer. Red once coolant temperature runs too hot.                                                                                                                                                                                |
|  | <b>Low fuel</b>                  | A fuel pump. Amber once your tank drops below the game's own low-fuel threshold.                                                                                                                                                         |
|  | <b>Low AdBlue</b>                | A droplet. Amber once your AdBlue/DEF level runs low.                                                                                                                                                                                    |
|  | <b>Battery / charging</b>        | A battery with its two terminals. Red if the charging system fails or voltage drops critically - on a real truck this can mean losing electrical power entirely, which is why it's red rather than amber.                                |

## Speed, RPM, and the odometer

Both dials sweep like a real analog gauge rather than snapping to a new position. Speed's needle and digital readout turn green at or under the posted limit, amber up to 5 over, and red beyond that - there's no fixed color band painted on the dial face itself, since what counts as "safe" changes with every speed limit sign you pass. RPM works the other way: its dial has a fixed red zone painted onto the last stretch of the arc, because that

position is always correct once the dial is scaled to your engine's own redline, whatever RPM number that happens to be.

Directly under the speedo sits a six-digit mechanical-style odometer, ticking up in whole miles or kilometers as you drive - it flashes briefly each time a digit actually rolls over.

## Fuel & AdBlue

One gauge face, two readings. Fuel is a swept needle with a painted red reserve zone in the last eighth of the dial and its own low-fuel telltale near "E". AdBlue shares the same face as four discrete quarter-segments below the needle's pivot - the same segmented-bar convention real trucks use for DEF gauges - lighting up as soon as the level reaches into each new quarter, so it never looks empty until it actually is.

## Cruise control & gear

Cruise shows its set speed at all times, colored green while engaged and dim grey while off. The gear box shows D/N/P/R for an automatic; for a manual truck it shows the real gear number, with an L/H suffix on trucks that have a range switch. On an H-shifter gearbox TruckHub hasn't been calibrated for yet, it falls back to showing the raw shifter slot (e.g. **S4**) rather than a confidently wrong gear number - see Gearbox Calibration in Section 07.

## Job panel

While a job's active: pickup and drop-off (with company names), cargo and its weight, payout, distance, and time left. Distance counts **up** from zero as you actually drive the route, not down from the job board's quoted estimate - a ferry crossing, a detour, or a recalculated route can all make the real distance longer or shorter than what you were quoted, and this stays accurate either way.

Below that: **Trip Needs**, the percentage of your current fuel this job will actually take to finish (red if you're short), and **Economy**, your rolling fuel consumption in MPG or L/100km depending on your unit settings.

## Rest timer

Green with over an hour left, amber under an hour, red at or past due. This row only appears if you've told TruckHub you have fatigue simulation switched on in-game (Section 07) - the game reports the same placeholder countdown whether the simulation is on or simply hasn't started ticking yet, so TruckHub has no way to tell those two states apart on its own.

## SECTION 03

# Color & Symbol Legend

One page, for whenever a light comes on and you can't remember what it means.

|                                                                                   |                 |                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Green</b>    | Normal, safe, or a driving aid working as intended (cruise, retarder, engine brake, turn signals, low beam, sidelights).                                                               |
|  | <b>Amber</b>    | Needs attention soon, or a deliberate state that changes handling (diff lock, lift axle up, low fuel/AdBlue, beacon, near-due rest timer).                                             |
|  | <b>Red</b>      | Safety-critical fault or danger: low air pressure, oil pressure, overheating, dead battery/charging failure, over the speed limit by more than 5, an axle unexpectedly off the ground. |
|  | <b>Blue</b>     | High beam, specifically - the one near-universal dashboard exception to the green/amber/red scheme.                                                                                    |
|  | <b>Dim grey</b> | Off, disengaged, or not applicable right now.                                                                                                                                          |
|  | <b>White</b>    | A neutral reading with no state to flag - a plain number or label.                                                                                                                     |

## Why the colors sometimes mean different things

A couple of icons bend the rule above on purpose, matching how the real vehicle they represent actually behaves: the retarder and engine brake glow **green** when engaged, not amber, because using them is a deliberate braking choice working exactly as intended - the same category as cruise control, not a fault. The differential lock and lift axle glow **amber** when active even though nothing is wrong, because both genuinely change how the truck handles and are worth being aware of while they're on.

## Suspension / axle labels (EXT window)

|                      |                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>W1, W2, W3...</b> | An ordinary steer or drive wheel - just numbered in order.                                                                                 |
| <b>LIFT</b>          | A liftable axle (a "tag" or "pusher", depending where it sits) - read directly from the game's own axle data, not guessed by position.     |
| <b>TAG</b>           | A fixed auxiliary axle - neither driven, steered, nor liftable - carrying extra load but never raised.                                     |
| <b>UP (amber)</b>    | A lift axle currently raised. Its suspension bar goes empty on purpose - there's nothing meaningful to read with the wheel off the ground. |

## SECTION 04

# The GPS Map

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Opened from the **GPS** tab. This is a real reconstruction of whichever game is running - the full ATS road network, or ETS2's, including the UK (reachable in-game only by ferry, and included on the map as real, drivable roads even though nothing connects it to the continent by land) - not a generic map with your position pasted on top. Your truck's position and heading update live, and once a job's destination is known, TruckHub computes a route that actually follows real roads to get there.

The route is calculated once, when the job starts (or when you open the map with one already active) - it isn't live-rerouted if you go off course, the same way most dedicated GPS units only recalculate when you ask them to or drift far enough away.

This includes ferry and train crossings (the UK to mainland Europe, most notably) - the route carries straight across the water as part of the same job, the same way it treats any other stretch of the trip.

## On the map

### Highway shields

Interstate, US Route, and State Route signs drawn in their real MUTCD shapes and colors, in ATS. ETS2 gets real European road signs too - green plates for E-roads, national motorways, and expressways; blue for national and regional roads - matching how the real signs look rather than a US-style shape.

### State / country names

Muted, letter-spaced capitals in the background - context, not something competing for attention with cities.

### Company names & logos

Every delivery company's marker shows its real in-game name and logo, sourced directly from the game's own files - both ATS and ETS2.

### City & town names

Sized and brightened by importance - major cities render larger and whiter, small towns smaller and dimmer.

### Facility markers

Real pictograms rather than letters - see below.

## Facility markers



A pump icon - only genuine, truck-usable gas stations. Decorative AI-traffic-only stations are filtered out.

|                                                                                   |                      |                                                                   |
|-----------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------|
|  | <b>Service</b>       | A wrench icon, for repair shops and truck dealers.                |
|  | <b>Rest stop</b>     | A bed icon, for parking areas you can actually stop and sleep at. |
|  | <b>Weigh station</b> | A plain “W” badge.                                                |

The map window is fully independent and resizable, with its own minimize, maximize, and reset-to-default-size controls in the title bar.

## Next-turn sign

A green advance-guide-sign - the same style highway exit and turn signs actually use - sits in the map's bottom-right corner while a route is active, showing which way the next turn goes and how far away it is. Toggle it off from the **Turns** button in the map's toolbar if you'd rather not have it. ATS only for now.

## LAN Mode

A **LAN** button sits in the map's own toolbar, alongside Pin and Clear. Turn it on and this same live map - your position, heading, and route - becomes viewable from another device on your home network through a plain browser: a phone or tablet propped up on the dash, for instance. Nothing needs installing on that other device.

Turning it on asks you to choose **No SSL** or **Use SSL**. Pick No SSL unless you have a specific reason not to - it's the recommended, works-everywhere option, and the only one that's reliable on an iPhone or iPad (Safari's TLS stack doesn't get along with the self-signed certificate the SSL option needs here). Once it's on, the address to type into the other device's browser is shown right there in the corner of the map.

**NOTE** The remote view is read-only and local-network-only. You can watch the live position and route from another device, but placing or clearing a pin, and turning LAN Mode itself on or off, only ever happens from the PC's own map window - and it only reaches devices on the same network, not the internet.

## SECTION 05

## Extended Instruments (EXT)

Everything the main dashboard doesn't have room for.

Opened from the **EXT** tab. Same rule as the main card: nothing shows until a supported game connects. Every element here can be switched on or off individually from this window's own settings button, separately from the main dashboard's toggles.

### The six gauges

|                    |                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oil Pressure       | 0–100 psi.                                                                                                                                                                                                                  |
| Oil Temperature    | Green / amber / red as it climbs toward the high end of the dial.                                                                                                                                                           |
| Water Temperature  | Red once the game's own overheat warning trips.                                                                                                                                                                             |
| Battery            | Scaled to your truck's own electrical system, not a fixed number - ATS trucks generally run 12V, ETS2 trucks 24V, and the dial always reads correctly for whichever you're driving.                                         |
| Brake Air Pressure | 0–150 psi, red once the game's own low-air warning trips.                                                                                                                                                                   |
| Brake Temperature  | Green under 260°C, amber 260–343°C, red above 343°C - real heavy-truck drum-brake service limits. A long descent can genuinely run brakes into the amber zone; sustained heat past the red line risks fade and drum damage. |

### Suspension deflection

One segmented bar per wheel, truck and trailer shown separately - filling from the bottom in six steps, the same segmented-cell look as the main dashboard's AdBlue gauge. It's the closest reading the game exposes to "how loaded is this axle" (more sag reads as more segments lit), not a calibrated weight scale.

Wheels are only ever shown for axles your specific truck or trailer actually has - there's no padding out to a generic maximum wheel count. See Section 03 for what the LIFT / TAG / UP labels mean.

### Trailer axle status

Per-wheel state for the attached trailer: **DOWN** (on the ground, normal), **LIFTED** (deliberately raised), or **AIRBORNE** (unexpectedly off the ground - worth a look). Beneath the wheel list, the tractor's own lift-axle control lamp shows whether you've commanded the trailer's axle up or down from the cab.

### Accessory wear

Engine, transmission, cabin, chassis, and wheel wear for the truck; body, cargo, chassis, and wheel wear for the trailer. Green under 50% worn, amber 50–80%, red beyond that.

## SECTION 06

# Job Log

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Opened from the **LOG** tab, which drops down a drawer beside the main card. Click the pop-out button in its corner to move the log into its own separate, resizable window if you'd rather keep it away from the main dashboard.

Every job you deliver is added to the top of the list with its pickup, drop-off, payout, and distance, alongside a running **Total Payout** and **Total Mileage** for the session. A cancelled or failed job is simply dropped, not logged.

**NOTE** The log is a running tally for this session only - it isn't saved to disk, and starts empty again the next time TruckHub opens.

## SECTION 07

# Settings

Two tabs: App Configure and UI Setup.

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## App Configure

**Updates** - check on demand, or let TruckHub check quietly in the background. A found update shows a one-click download link here, and puts a small amber glow on the Settings button on the main dashboard until you've opened Settings and seen it.

**Game install & mod folder locations** - TruckHub auto-detects standard Steam installs for both games. If a copy isn't found (a non-Steam install, or a library drive Steam isn't currently seeing), point it at the right folder manually here. The mod folder path is what the Mod Order tool (Section 08) scans.

## Gearbox calibration

American Truck Simulator resets an H-shifter to neutral the instant the game window loses focus - so tabbing out to read the current gear doesn't work, the game's own telemetry genuinely reports neutral the moment you do it. The workaround:

1. While driving, with the game still focused, shift into the gear you want to label.
2. Press **Ctrl+Alt+G** to capture that shifter position.
3. Come back to Settings (whenever's convenient - the capture is remembered) and type in which gear that actually was, then press Set gear.

Low/High range is detected and labeled automatically once you've calibrated a position. A splitter that reuses the same physical stick slots for different gears needs its own separate capture and Set, using the splitter-specific row that appears once TruckHub detects your gearbox has one.

## UI Setup

Unit toggles for speed, cargo weight, and fuel volume. Below that, a full on/off checklist for every component on the main dashboard - current speed, odometer, speed limit sign, clock, cruise control, parking brake, lights row, gear/RPM, route, cargo, payout, distance, time left, fuel, trip fuel forecast, fuel economy, AdBlue, and the warning cluster. Turning something off shrinks the card to fit - nothing leaves a blank gap behind.

**Fatigue simulation** - a single checkbox telling TruckHub whether you've enabled the game's own fatigue simulation. The game reports the same countdown value whether the simulation is on or simply hasn't started counting yet, so TruckHub can't detect this on its own; the Rest timer row only appears once you've confirmed it here.

## SECTION 08

# Mod Load Order

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Opened with the Mod Order button on the main dashboard's title bar. TruckHub scans your installed mods for each game and suggests an order to load them in, based on community-recommended category tiers (map mods, vehicle mods, sound mods, and so on, each in the order that tends to cause the fewest conflicts).

This is a reference list only. TruckHub never reorders, enables, or disables anything in your Mod Manager - you read the suggested order here and apply it yourself.

**IMPORTANT** This is the community's best-effort order, not a guarantee. Some mods are fussy about load order regardless of category, and can still conflict or crash even when correctly ordered. Treat it as a strong starting point, not a fix for every problem.

## SECTION 09

# Quick Reference

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## The one hotkey worth remembering

**Ctrl + Alt + G**      Capture the current H-shifter slot for gearbox calibration (Section 07). Must be pressed while the game window is focused and you're actually in the gear you want to label.

## Color legend, condensed

|                                                                                     |                 |                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Green</b>    | Normal, safe, or a driving aid working as intended (cruise, retarder, engine brake, turn signals, low beam, sidelights).                                                               |
|    | <b>Amber</b>    | Needs attention soon, or a deliberate state that changes handling (diff lock, lift axle up, low fuel/AdBlue, beacon, near-due rest timer).                                             |
|    | <b>Red</b>      | Safety-critical fault or danger: low air pressure, oil pressure, overheating, dead battery/charging failure, over the speed limit by more than 5, an axle unexpectedly off the ground. |
|   | <b>Blue</b>     | High beam, specifically - the one near-universal dashboard exception to the green/amber/red scheme.                                                                                    |
|  | <b>Dim grey</b> | Off, disengaged, or not applicable right now.                                                                                                                                          |
|  | <b>White</b>    | A neutral reading with no state to flag - a plain number or label.                                                                                                                     |

TruckHub reads game telemetry only - it cannot see or modify anything else on your system, and it doesn't send anything anywhere. If something on the HUD ever looks wrong, the game's own dashboard is always the source of truth: TruckHub is a second window onto the same numbers, not a separate calculation.